

A CONCEPT IMPROVEMENT DESIGN OF THE GIRDER ADJUSTMENT SYSTEM FOR TPS STORAGE RING

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Abstract

The girder adjustment system of TPS storage ring can fine adjust each girder in 6 axes with 6 kinematic mounting motorized cam movers. The installation of the TPS had demonstrated this design. However, this design is freely mounted with gravity and the 1st natural frequency is less than 30Hz even with supplement side locking system. Moreover, the motor controller restricts the beginning power output and sometimes the girder will fall down when the electromagnetic motor brake is released.

A concept improvement design is thus introduced to modify these situations. In this design, a worm gearbox addition can raise the reduction ratio to prevent the falling and inverse kinematic mounting movers with strong springs not only firmly lack the girder to raise the natural frequency but also preserve the motorized algorithm. This paper describes the design in detail.

INTRODUCTION

Taiwan Photon Source (TPS) was finished installation at 2014 and reach the design spec (500mA) at the end of 2015. The girder system for the TPS storage ring is of an auto-tuning design in order to align the girders precisely and quickly with less manpower. Each girder can be fine adjusted in 6 axes with 6 motorized cam movers of kinematic mounting design on 3 pedestals. With sensors between each girder, there are 72 girders to make up a whole ring auto-alignment girder system [1-5]. There had 3 auto-tunings been performed before commissioning. In the design spec, the 1st natural frequency more than 30 Hz of the girder system was requested. But after installed with all the components, it can be measured to about 24~25 Hz and 26~28 Hz with the side locking systems.

After a few years operation, the tunnel was found some local settlement areas from yearly survey works as shown in the Fig. 1. So two locally girder system adjustments were performed in 2020 & 2021 to correct this situation [6]. However, the motor controller restricts the beginning power output and sometimes the girder will fall down to the hard stop when the electro-magnetic motor brake is released.

Recently, the study of TPS upgrade was initialized after the 3rd beamline construction is nearly finished. A concept improvement girder design is thus introduced to modify the weak situations found. If the formal study begins, a prototype may be setup with the existing backup girder system in the laboratory.

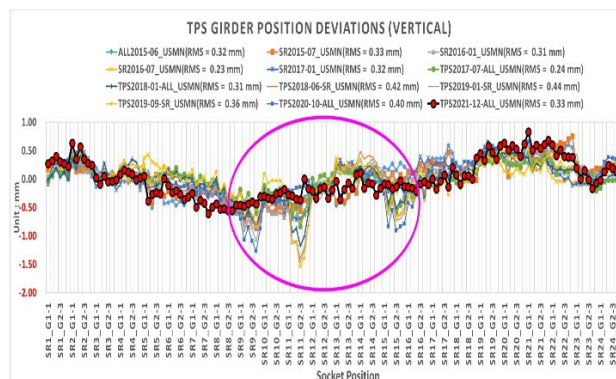


Figure 1: TPS girder position deviation from 2015~2021.

GIRDER SYSTEM DESIGN

The single girder system of the TPS storage ring is shown as in the Fig. 2. This design is freely mounted with gravity and the 1st natural frequency is less than 30 Hz even with supplement electrical side locking system. An improvement design is introduced as in the Fig. 3.

In this design, the cam movers and transfer unit housings will be installed upside down. There are mold springs inside the 3 pedestals. These mold springs not only support the total weight of girder system (16 tons) but also will be suppressed with the movers with another 12~16 tons when adjustment.

The girder system will be strongly and fully kinematically clamped with the cam movers and the mold springs. The hard stopper and locking system in the original design will be no longer need and the 1st natural frequency is expected to be raised at least $\sqrt{2}$ times to more than 36 Hz. A detailed components design and simulation will be performed when the formal study is assigned.

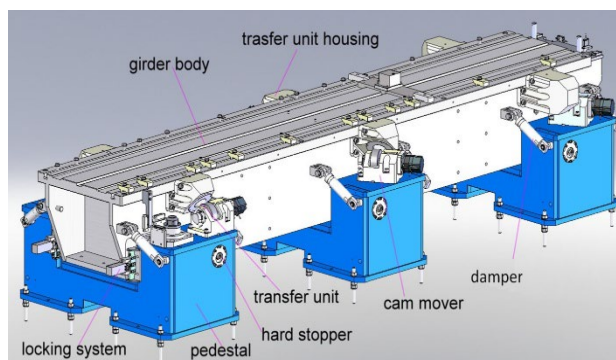


Figure 2: A single girder system of the TPS storage ring.

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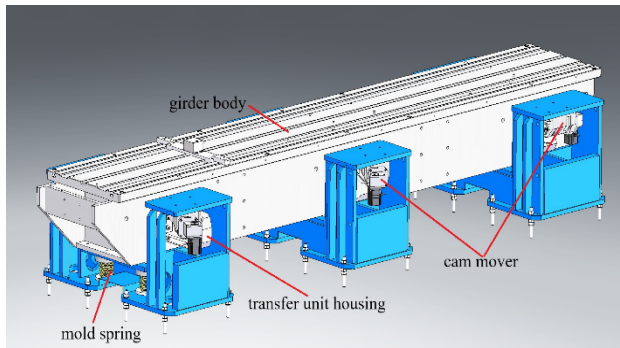


Figure 3: An improvement girder system design.

ADJUSTING RANGE

Since this improvement design is also kinematic mounting type, the motorized algorithm can be preserved and resumed. Two coordinate systems are to be established at girders and movers separately, from the rigid body assumption, the adjustment of center position of each ball can be calculated as in Fig. 4.

Due to the symmetric kinematic V groove type arrangement (120 degree), the girder moving range in horizontal direction is only 1/2 of the mover range in transverse (X) direction (± 4.5 mm) and $\sqrt{3}/2$ in longitudinal (Z) direction (± 7 mm), as in Fig. 5.

A modification is to adopt unsymmetric V groove design as in Fig. 6. A 135 degree arrangement can make the mover range in transverse (X) direction and longitudinal (Z) direction same of $1/\sqrt{2}$ (± 6.3 mm). An extreme arrangement is the perpendicular type to preserve the full range adjustment with 2 more redundant movers (8 movers) and increase the system stiffness furtherly if longer girder will be adopted.

The adjustment range of three types can be listed as in the Table 1.

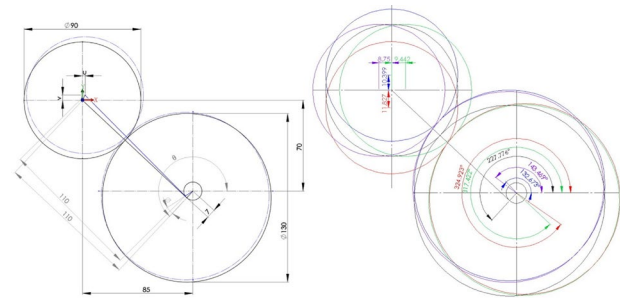


Figure 4: Adjusting range of the mover with transfer unit ball pair.

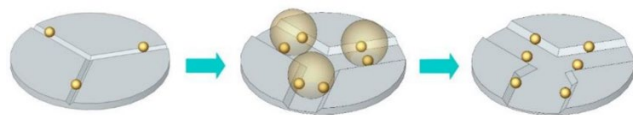


Figure 5: Expand 3-Groove type kinematic mounting from 3 balls to 6 balls.

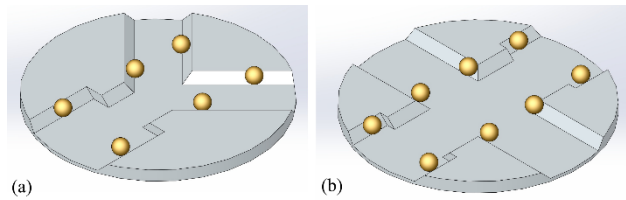


Figure 6: (a) 135-degree unsymmetric V groove design. (b) perpendicular type with 2 more redundant movers (8 movers).

Table 1: Three Mover Arrangement Types Adjustment Range

	Symmetric [mm]	Unsymmet- ric[mm]	Perpen- dicular [mm]
Transverse (X)	± 4.5	± 6.3	± 8.8
Vertical (Y)	± 10.4	± 10.4	± 10.4
Longitudinal (Z)	± 7	± 6.3	± 8.8

MOVER IMPROVEMENT

The motorize mover system is assembled with a stepping motor and a 160-reduction ration harmonic gear box. The stepping motor includes a normal close electro-magnetic brake. Since the motor controller restricts the beginning power output and the control system has a delayed release time design. However, sometimes the girder will drop down to the hard stop when released as in the local settlement re-adjustment. So, an improvement design with worm gear to more 10 times reduction will be adopted. In this design, a small motor could be used and self-lock with high ratio reduction friction is expected as in Fig. 7.

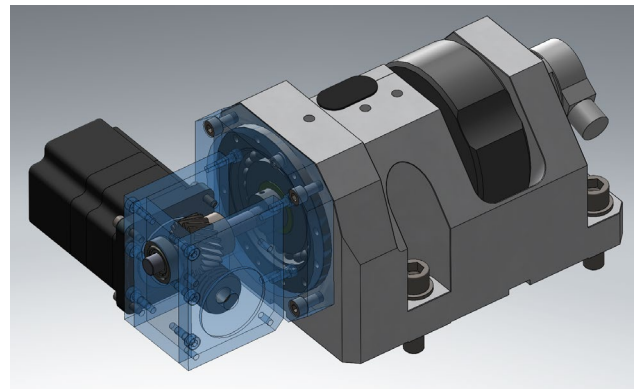


Figure 7: Mover modification design.

CONCLUSION

The motorized girder system of TPS storage ring had demonstrated the easy and precise adjustment ability with stability during the commissioning and decade operation. However, the reliability is not quit absolute during read-

justment and further hard stopper had added to the girder system due to restricted beginning power output of the motor controller.

A concept improvement girder design is thus introduced to modify the weak situations found. If the formal study begins, a further system design and simulation will be performed. A prototype may be setup with the existing backup girder system in the laboratory.

REFERENCES

- [1] T. C. Tseng *et al.*, “A precise 6-axies girder system with cam mover mechanism”, presented at MEDSI'06, Hyogo, Japan, May 2006.
- [2] T. C. Tseng *et al.*, “Design and prototype testing of the girder system for TPS”, presented at SRI'08, Saskatoon, Canada, Jun. 2008.
- [3] T. C. Tseng *et al.*, “From Survey Alignment toward Auto-alignment for the Installation of the TPS Storage Ring Girder System”, in *Proc. IPAC'11*, San Sebastian, Spain, Sep. 2011, paper MOPO034, pp. 559-561.
- [4] T.C. Tseng *et al.*, “Installation And Implementation of an Auto-Alignment Girder System for TPS Storage Ring”, presented at MEDSI'14, Melbourne, Australia, Oct. 2014.
- [5] T. C. Tseng *et al.*, “The Auto-Alignment Girder System of TPS Storage Ring”, in *Proc. IPAC'15*, Richmond, VA, USA, May 2015, pp. 3649-3653.
[doi:10.18429/JACoW-IPAC2015-THYB2](https://doi.org/10.18429/JACoW-IPAC2015-THYB2)
- [6] T. C. Tseng *et al.*, “An IR Photonic Type HLS system developed at NSRRC”, presented at International Workshop on Accelerator Alignment (IWAA), Menlo Park, CA, USA, Oct. 2024.